Tommaso de Martino

+39 333 847 9300 | demar.tommaso@gmail.com | linkedin.com/in/tommaso-de-martino | tdistudent27.github.io

WORK EXPERIENCE

European Central Bank – Statistics and Data Science Trainee *Feb 2026 – Present*
Frankfurt, Germany

- Built a factor-based macro forecasting framework using 126 U.S. economic time series (1959–2025), extracting 7 latent factors to produce rolling real-time forecasts for industrial production and CPI inflation, achieving out-of-sample RMSEs of 1.37 and 0.60, respectively
- Led the development of Python-based pipelines automating market and credit risk analytics across 100+ European financial institutions, processing 12 million data points and reducing processing time from several days to minutes
- Partnered with risk officers to design and implement quantitative frameworks for rating, liquidity and asset-risk monitoring, taking requirements from problem definition to production and supporting ongoing supervisory reporting across European banks
- Validated and reconstructed 10,000 weekly equity and FX market observations using Bloomberg data, recovering missing values from daily series and performing valuation consistency checks that increased valid data coverage by 25%

EDUCATION

MS Economics and Finance 110/110 cum laude – Sapienza University of Rome *Sep 2023 – Oct 2025*

- Relevant coursework: Quantitative Methods, Financial Econometrics, Machine Learning in Finance, Mathematics
- Merit scholarship: tuition fee waiver awarded to top 5% of cohort based on GPA
- Thesis: Asset Pricing with Machine Learning

BS Economics – Sapienza University of Rome *Sep 2020 – Oct 2023*

- Relevant coursework: Economic Theory, Monetary Policy, Financial Mathematics, Statistics
- Thesis: Dedollarization of the International Monetary System

PROJECTS

Asset Pricing with Machine Learning

- Developed an out-of-sample machine learning approach to forecast cross-sectional stock returns across 3 milion firm-month observations, 23,729 U.S. firms and 807 predictors, with neural networks delivering the strongest predictive performance
- Translated model forecasts into long-short equity portfolios, with the best strategy delivering a 38.11% annualized return and a Sharpe ratio of 1.98

U.S. Treasury Yield Curve Modeling and Forecasting

- Modeled the U.S. Treasury yield curve using the Diebold-Li (Nelson-Siegel) approach across 8 maturities (3M-10Y), fitting 437 monthly observations with a 6bps in-sample RMSE
- Forecasted level, slope and curvature over 1/3/6/12-month horizons using AR and VAR models with expanding-window out-of-sample validation, finding significant long-horizon outperformance versus a random walk for several maturities

Volatility Forecasting – Euro Stoxx 50

- Forecasted Euro Stoxx 50 volatility using market signals across FX, rates, equity indices and commodities over 3,763 daily observations
- Benchmarked GARCH against neural network and Random Forest models, with the best neural network reducing out-of-sample MSE by 20.4% and RMSE by 10.8% relative to GARCH

Systematic FX Strategy Research and Validation

- Developed a EUR/USD backtesting engine using 4-hour trading signals and 1-minute data to model execution, incorporating transaction costs, overnight carry and risk-based position sizing
- Stress-tested 972 parameter configurations across 1,000 randomized return series to assess overfitting risk, with only 1.1% matching or exceeding the strategy's performance

SKILLS & INTERESTS

Technical

- Python (advanced), Pandas, Polars, NumPy, scikit-learn, PyTorch, R (advanced), SQL, Excel (advanced), Git and GitHub for version control and collaborative development
- Time-series modelling, yield curve modelling, backtesting, portfolio construction, machine learning, market data analysis, repo and securities financing mechanics

Languages: Italian (native), English (fluent), Spanish (fluent), French (intermediate)

Interests: Financial markets, Muay Thai, nature